SUPPLEMENTARY MATERIALS FOR "ADAPTIVE BAYESIAN MULTIVARIATE SPLINE KNOT INFERENCE WITH PRIOR SPECIFICATIONS ON MODEL COMPLEXITY"

Junhui He
Department of Mathematical Sciences
Tsinghua University
Beijing, China

Ying Yang
Department of Statistics and Data Science
Tsinghua University
Beijing, China
yangying@tsinghua.edu.cn

Jian Kang
Department of Biostatistics
University of Michigan, Ann Arbor
Michigan, United States
jiankang@umich.edu

A Web Appendix A

A.1 Proof of Lemma 1

Proof. The marginal likelihood $p(y|k,\xi)$ is given by

$$p(y|k,\xi) = p(y|Z) = \int_{(0,\infty)} \int_{\mathbb{R}^\nu} p(y|Z,\beta,\sigma)\pi(\beta|Z,\sigma)\pi(\sigma)d\beta d\sigma.$$

According to the specified priors, we have

$$p(y|Z,\beta,\sigma) = \frac{1}{(2\pi\sigma^2)^{m/2}} \exp\{-\frac{1}{2\sigma^2}(y - Z\beta)^\top (y - Z\beta)\},$$
$$\pi(\beta|Z,\sigma) = \frac{1}{(2\pi m\sigma^2)^{\nu/2}} |Z^\top Z|^{1/2} \exp\{-\frac{1}{2m\sigma^2}\beta^\top Z^\top Z\beta\}.$$

Then the Fubini's Theorem (Royden and Fitzpatrick, 2010) implies that,

$$p(y|k,\xi) = \int_0^\infty \frac{1}{(2\pi\sigma^2)^{m/2}(m+1)^{\nu/2}} \exp\{-\frac{1}{2\sigma^2}a_{k,\xi}\}\pi(\sigma)d\sigma,$$

where $a_{k,\xi} = y^\top (I_m - \frac{m}{m+1} Z(Z^\top Z)^{-1} Z^\top) y$. With change of variables $w = \sigma / \sqrt{a_{k,\xi}}$, we have $p(y|k,\xi) \propto (m+1)^{-\nu/2} a_{k,\xi}^{-m/2}$. It follows from $\pi(k,\xi) \propto \tau(\mathcal{M}_k)^{-\gamma}$ that $p(k,\xi|y) \propto (m+1)^{-\nu/2} a_{k,\xi}^{-m/2} \tau(\mathcal{M}_k)^{-\gamma}$. □

A.2 Proof of Lemma 3

Proof. According to the detailed balance equation,

$$\alpha(k',\xi'|k,\xi) = \min\left\{1,\ \frac{p(k',\xi'|y) q(k,\xi|k',\xi')}{p(k,\xi|y) q(k',\xi'|k,\xi)}\right\},$$

where $p(k',\xi'|y)$ and $p(k,\xi|y)$ are the posterior densities, and $q(k,\xi|k',\xi')$ and $q(k',\xi'|k,\xi)$ are the proposal distributions. We claim that under our prior and proposal specification, the relation $\pi(k,\xi) q(k',\xi'|k,\xi) = \pi(k',\xi') q(k,\xi|k',\xi')$ holds. Without loss of generality, assume that $d = 1$. From the prior specification,

$$\pi(k,\xi) = \pi(\xi|k)\pi(k) \propto \tau(\mathcal{M}_k)^{-\gamma}, \quad \pi(k',\xi') \propto \tau(\mathcal{M}_{k'})^{-\gamma}.$$

1. Birth step ($k' = k+1$): The prior ratio is

$$\frac{\pi(k+1,\xi')}{\pi(k,\xi)} = \left(\frac{\tau(\mathcal{M}_{k+1})}{\tau(\mathcal{M}_k)}\right)^{-\gamma} = \left(\frac{n-k}{k+1}\right)^{-\gamma}.$$

   The proposal ratio is

$$\frac{q(k+1,\xi'|k,\xi)}{q(k,\xi|k+1,\xi')} = \frac{b_k(k+1)}{d_{k+1}(n-k)} = \frac{\pi(k+1)(k+1)}{\pi(k)(n-k)} = \left(\frac{n-k}{k+1}\right)^{-\gamma}.$$

   The second equality uses the identity $\pi(k) b_k = \pi(k+1) d_{k+1}$. Therefore, we have $\pi(k,\xi) q(k+1,\xi'|k,\xi) = \pi(k+1,\xi') q(k,\xi|k+1,\xi')$.

2. Death step ($k' = k-1$): The derivation mirrors the Birth case, with the numerator and denominator reversed.

3. Relocation step ($k' = k$): Here,

$$\frac{\pi(k,\xi')}{\pi(k,\xi)} = 1, \quad \frac{q(k,\xi'|k,\xi)}{q(k,\xi|k,\xi')} = 1.$$

Since $p(k,\xi|y) \propto p(y|k,\xi)\pi(k,\xi)$, we have

$$\alpha(k',\xi'|k,\xi) = \min\left\{1,\ \frac{p(y|k',\xi')}{p(y|k,\xi)}\right\}.$$

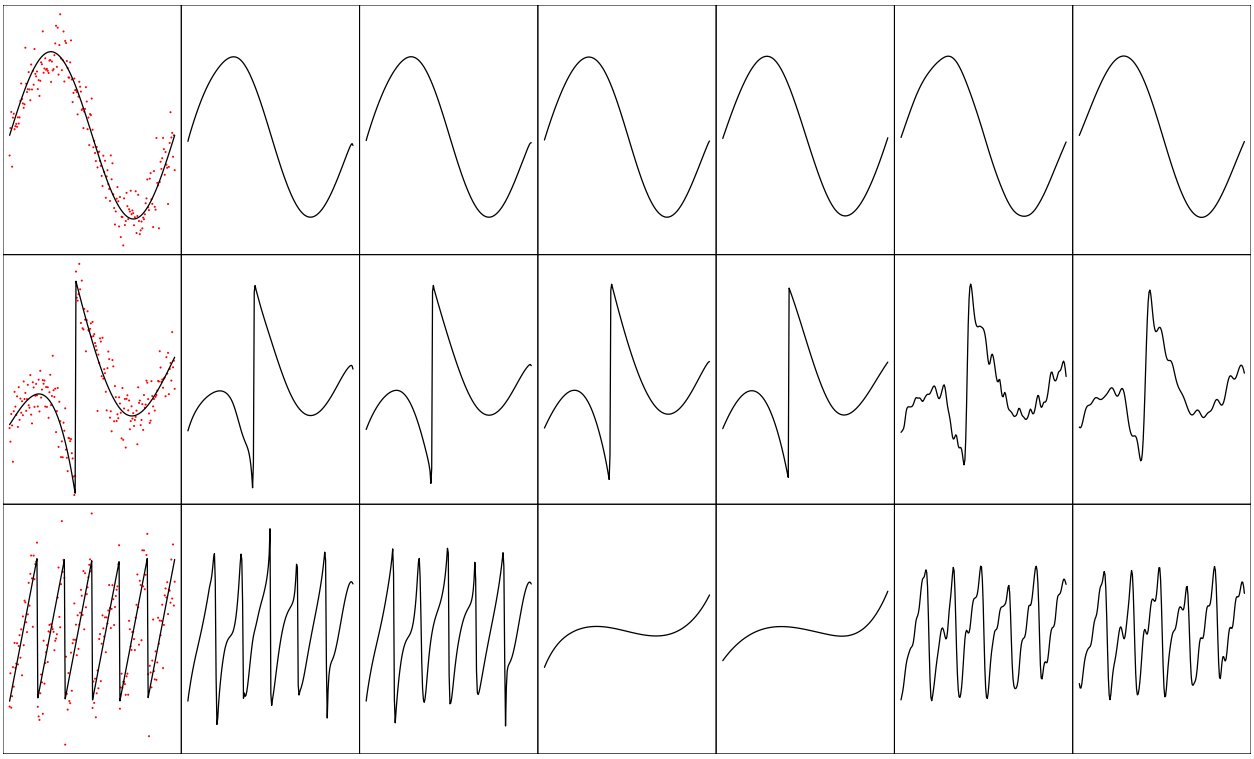
Thus, Lemma 1 implies the acceptance probability in RJMCMC. For the EBIC-based approximation, we substitute $\hat{p}$ for the corresponding p. □

B Web Appendix B

For curve fitting, the proposed method is benchmarked against Bayesian Adaptive Knot Regression Splines (BARS) (Dimatteo et al., 2001), Smoothing Splines (SMS) (Green and Silverman, 1994), and Gaussian Process Regression (GPR) (Rasmussen and Williams, 2005). BARS is a Bayesian method for knot estimation in univariate splines, which employs a Poisson prior for the number of knots. However, this method cannot be directly extended to multivariate splines. SMS represents a class of penalized splines that incorporate a roughness penalty. It models the unknown function as a spline with a large number of knots, enabling a high degree of flexibility, while the roughness penalty regulates the smoothness of the spline. SMS is implemented using the build-in funtion `smooth.spline`, and GPR with the squared exponential kernel is implemented using the `R` package `GauPro` (Erickson, 2025).

We examine three curves with varying levels of smoothness, labeled as Curves 1–3. The output noise follows a Gaussian distribution with standard deviations of 2, 4, and 1 for Curves 1, 2, and 3, respectively. Web Figure 1 depicts the curves along with the training samples. Curve 1 is continuous, whereas Curves 2 and 3 exhibit discontinuities with one or more jump breakpoints.

To investigate the influence of the parameter γ in the proposed method, we consider three scenarios where γ is set to 0.8, 1.0, and 1.2. All methods are evaluated under sample sizes $m = 200$ and 500, with the experiment repeated 50 times. Prediction performance is assessed using the Root Mean Squared Error (RMSE).



Web Figure 1: Curve fitting with sample size $m = 200$. Rows correspond to Curves 1–3 (top to bottom). Columns correspond to the ground truth with noisy samples (red points), followed by estimates using ABMRS (with $\gamma \in \{0.8, 1.0, 1.2\}$), BARS, SMS, and GPR (left to right).

The fitted curves are shown in Columns 2–6 of Web Figure 1, and the RMSEs are summarized in Web Table 1. For the smooth curve (Curve 1), all methods achieve low RMSEs. However, for the discontinuous curves (Curves 2 and 3), SMS and GPR fail to capture the jump discontinuities, resulting in substantially degraded performance. The BARS method performs well when the true number of knots is relatively small. However, for highly oscillatory true functions (Curve 3), BARS tends to include too few knots in its estimation. In this example, the proposed method with $\gamma = 1.2$ yields a high RMSE for Curve 3 when the sample size is $m = 200$, indicating that $\gamma > 1$ can lead to overly regularized models for highly oscillatory curves. This issue can be substantially alleviated by choosing a smaller value of γ or increasing the sample size.

Web Table 1: RMSE based on 50 repeated experiments in Curves 1–3 with sample size $m = 200$ and 500. Curve 1 is smooth, while Curves 2 and 3 exhibit jump discontinuities.

	m	Curve 1	Curve 2	Curve 3
ABMRS ($\gamma = 0.8$)	200	0.378 ± 0.092	1.385 ± 0.574	$\mathbf{0.680 \pm 0.143}$
ABMRS ($\gamma = 1.0$)		0.391 ± 0.099	1.328 ± 0.610	0.751 ± 0.278
ABMRS ($\gamma = 1.2$)		0.430 ± 0.122	1.314 ± 0.601	1.533 ± 0.496
BARS		0.411 ± 0.119	$\mathbf{1.290 \pm 0.839}$	1.742 ± 0.202
SMS		$\mathbf{0.364 \pm 0.100}$	2.838 ± 0.542	0.806 ± 0.110
GPR		0.365 ± 0.261	2.999 ± 0.547	0.936 ± 0.356
ABMRS ($\gamma = 0.8$)	500	0.245 ± 0.069	0.886 ± 0.368	$\mathbf{0.425 \pm 0.058}$
ABMRS ($\gamma = 1.0$)		0.242 ± 0.068	0.918 ± 0.356	0.437 ± 0.061
ABMRS ($\gamma = 1.2$)		0.242 ± 0.072	0.957 ± 0.359	0.457 ± 0.064
BARS		0.288 ± 0.063	$\mathbf{0.869 \pm 0.443}$	0.615 ± 0.364
SMS		0.249 ± 0.071	2.254 ± 0.339	0.619 ± 0.052
GPR		$\mathbf{0.202 \pm 0.065}$	2.511 ± 0.868	0.713 ± 0.301

C Web Appendix C

To further evaluate model performance, we analyze five additional surface functions given by Denison et al. (1998), specified as follows:

1. Case 1: $g_1(x) = 10.391[(x_1 - 0.4)(x_2 - 0.6) + 0.36]$;
2. Case 2: $g_2(x) = 24.234[r^2(0.75 - r^2)]$, where $r^2 = (x_1 - 0.5)^2 + (x_2 - 0.5)^2$;
3. Case 3: $g_3(x) = 42.659[0.1 + \hat{x}_1(0.05 + \hat{x}_1^4 - 10\hat{x}_1^2\hat{x}_2^2 + 5\hat{x}_2^4)]$, where $\hat{x}_1 = x_1 - 0.5$, $\hat{x}_2 = x_2 - 0.5$;
4. Case 4: $g_4(x) = 1.3356\Big\{1.5(1 - x_1) + \mathrm{e}^{2x_1 - 1}\sin[3\pi(x_1 - 0.6)^2] + \mathrm{e}^{3(x_2 - 0.5)}\sin[4\pi(x_2 - 0.9)^2]\Big\}$;
5. Case 5: $g_5(x) = 1.9\Big\{1.35 + \mathrm{e}^{x_1}\sin[13(x_1 - 0.6)^2] \times \mathrm{e}^{-x_2}\sin(7x_2)\Big\}$.

The output noise for each surface follows a Gaussian distribution with standard deviations of 0.25, 0.22, 0.22, 0.25 and 0.25, respectively.

We compare the proposed method with Thin Plate Regression Splines (TPRS) (Wood, 2003), Tensor Product Regression Splines (TEPS) (Wood, 2017), Bayesian MARS (BMARS) (Denison et al., 1998), and Gaussian Process Regression (GPR) (Rasmussen and Williams, 2005). Both TPRS and TEPS extend smoothing splines to multivariate settings using prespecified knots. We implement TPRS and TEPS using the `R` package `mgcv` with a roughness penalty (Wood and Wood, 2015). BMARS is implemented using the `R` package `BASS` (Francom and Sansó, 2020), and GPR with the Matérn kernel is implemented using the `R` package `GauPro` (Erickson, 2025). For BMARS, we consider spline degrees 1, 2, and 3, and report the best results.

In this example, we employ cubic splines and consider $\gamma \in \{0.8, 1.0, 1.2\}$ in the proposed method. All methods are evaluated under sample sizes $m = 250$ and 1000, with the experiment repeated 50 times. Model performance is assessed using the Root Mean Squared Error (RMSE). To evaluate predictive uncertainty, we construct prediction intervals for the true functions and compare the average coverage and interval widths across methods.

Web Table 2 reports the results for the additional surface fitting problem. All methods achieve low RMSEs across all scenarios, with RMSEs decreasing as the sample size increases. When evaluating predictive uncertainty, all compared methods achieve an empirical coverage for their prediction intervals that matches the nominal confidence level. The credible or confidence intervals for the mean function—which exclude noise variance—tell a slightly different story. Here, our proposed method yields moderately lower coverage than TPRS, TEPS, and GPR. Even so, it stays within a reasonable range, unlike BMARS, which suffers from consistently poor coverage across every case. Finally, the proposed method proves highly robust to the choice of γ, showing very similar results across different values.

D Web Appendix D

Web Table 3 reports the additional results about predictive uncertainty in Section 4.2.1. When evaluating the coverage of 95% credible (confidence) intervals for the mean function, the proposed method achieves moderately lower, yet still reasonable, coverage compared to TPRS, TEPS, and GPR. This difference is expected: TPRS, TEPS and GPR rely on frequentist interval construction and therefore enjoy nominal frequency guarantees, whereas the proposed method is fully Bayesian and constructs confidence intervals using posterior quantiles, which do not provide exact frequentist coverage. BMARS, by contrast, attains consistently poor coverage across all surfaces.

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Web Table 2: Surface fitting results for Cases 1–5 with sample size $m = 250$ and 1000 under 50 repeated experiments.

(a) Root Mean Square Errors (RMSE) ($\times 10^{-1}$)

	m	Case 1	Case 2	Case 3	Case 4	Case 5
ABMRS ($\gamma = 0.8$)		0.67 ± 0.15	0.86 ± 0.16	1.02 ± 0.16	1.34 ± 0.21	1.55 ± 0.39
ABMRS ($\gamma = 1.0$)		0.67 ± 0.15	0.86 ± 0.16	$\mathbf{1.01 \pm 0.16}$	1.34 ± 0.21	1.56 ± 0.37
ABMRS ($\gamma = 1.2$)		0.67 ± 0.15	0.86 ± 0.16	1.01 ± 0.16	1.34 ± 0.21	1.56 ± 0.37
TPRS	250	0.86 ± 0.14	0.91 ± 0.13	2.08 ± 0.42	1.42 ± 0.15	1.52 ± 0.24
TEPS		$\mathbf{0.43 \pm 0.17}$	0.86 ± 0.13	1.45 ± 0.26	1.44 ± 0.27	1.44 ± 0.28
BMARS		0.51 ± 0.16	0.89 ± 0.17	1.24 ± 0.29	$\mathbf{0.78 \pm 0.13}$	2.66 ± 0.82
GPR		0.58 ± 0.11	$\mathbf{0.81 \pm 0.12}$	1.51 ± 0.35	1.14 ± 0.16	$\mathbf{1.36 \pm 0.19}$
ABMRS ($\gamma = 0.8$)		0.30 ± 0.05	0.43 ± 0.06	0.63 ± 0.08	0.70 ± 0.10	$\mathbf{0.68 \pm 0.08}$
ABMRS ($\gamma = 1.0$)		0.30 ± 0.05	0.43 ± 0.06	0.66 ± 0.06	0.72 ± 0.10	0.69 ± 0.09
ABMRS ($\gamma = 1.2$)		0.30 ± 0.05	0.43 ± 0.06	0.68 ± 0.06	0.71 ± 0.10	0.71 ± 0.09
TPRS	1000	0.47 ± 0.05	0.54 ± 0.06	0.97 ± 0.11	0.77 ± 0.06	0.79 ± 0.07
TEPS		$\mathbf{0.18 \pm 0.07}$	0.50 ± 0.06	0.71 ± 0.08	0.71 ± 0.06	0.76 ± 0.07
BMARS		0.27 ± 0.10	$\mathbf{0.39 \pm 0.08}$	$\mathbf{0.62 \pm 0.09}$	$\mathbf{0.45 \pm 0.09}$	1.13 ± 0.18
GPR		0.27 ± 0.05	0.45 ± 0.06	0.64 ± 0.07	0.60 ± 0.06	0.69 ± 0.06

(b) Average coverage and width of 95% prediction intervals

	m	Case 1	Case 2	Case 3	Case 4	Case 5
ABMRS ($\gamma = 0.8$)		93.8% (9.69)	93.4% (8.77)	93.9% (8.96)	93.5% (10.34)	93.0% (10.36)
ABMRS ($\gamma = 1.0$)		93.8% (9.69)	93.3% (8.78)	93.9% (8.96)	93.5% (10.33)	93.1% (10.40)
ABMRS ($\gamma = 1.2$)		93.8% (9.69)	93.4% (8.78)	93.9% (8.97)	93.5% (10.33)	93.0% (10.43)
TPRS	250	93.9% (10.03)	94.3% (9.12)	90.9% (9.47)	93.1% (10.62)	93.1% (10.47)
TEPS		94.3% (9.74)	94.7% (9.19)	94.1% (9.81)	94.5% (11.10)	94.4% (10.96)
BMARS		94.1% (9.79)	93.6% (9.40)	91.1% (11.10)	93.8% (10.39)	92.3% (14.45)
GPR		94.5% (9.88)	94.6% (9.11)	94.2% (10.30)	94.6% (10.71)	94.5% (10.86)
ABMRS ($\gamma = 0.8$)		94.8% (9.79)	94.5% (8.69)	94.6% (8.84)	94.6% (9.97)	94.5% (9.93)
ABMRS ($\gamma = 1.0$)		94.8% (9.79)	94.5% (8.69)	94.6% (8.88)	94.6% (10.01)	94.5% (9.93)
ABMRS ($\gamma = 1.2$)		94.8% (9.79)	94.5% (8.69)	94.7% (8.90)	94.6% (10.01)	94.6% (9.97)
TPRS	1000	94.9% (9.96)	94.9% (8.85)	94.7% (9.23)	94.8% (10.18)	94.7% (10.20)
TEPS		94.8% (9.80)	94.8% (8.83)	94.9% (9.04)	94.9% (10.17)	94.9% (10.19)
BMARS		94.7% (9.79)	94.0% (8.89)	93.6% (9.15)	94.3% (9.98)	94.0% (10.57)
GPR		94.9% (9.85)	94.8% (8.81)	94.9% (8.97)	95.0% (10.07)	94.9% (10.14)

* Values are reported as $a(b)$, where a denotes coverage and b denotes width ($\times 10^{-1}$).

(c) Average coverage and width of 95% credible or confidence intervals for the mean function

	m	Case 1	Case 2	Case 3	Case 4	Case 5
ABMRS ($\gamma = 0.8$)		93.6% (2.35)	92.7% (2.84)	84.6% (2.61)	84.8% (3.33)	85.0% (3.70)
ABMRS ($\gamma = 1.0$)		93.6% (2.35)	92.7% (2.85)	84.3% (2.60)	84.8% (3.30)	84.5% (3.66)
ABMRS ($\gamma = 1.2$)		93.6% (2.35)	92.6% (2.85)	83.8% (2.56)	84.4% (3.28)	84.1% (3.63)
TPRS	250	96.6% (3.40)	95.6% (3.23)	90.2% (5.05)	93.4% (4.99)	93.4% (5.07)
TEPS		95.1% (1.66)	96.0% (3.30)	96.1% (5.10)	96.6% (5.46)	95.4% (5.51)
BMARS		89.4% (1.14)	80.3% (2.19)	65.3% (2.08)	84.5% (2.16)	53.2% (3.22)
GPR		95.8% (2.24)	96.1% (3.00)	95.7% (4.89)	96.4% (4.56)	96.2% (5.41)
ABMRS ($\gamma = 0.8$)		95.8% (1.16)	90.6% (1.38)	76.3% (1.43)	84.5% (1.75)	86.4% (1.84)
ABMRS ($\gamma = 1.0$)		95.8% (1.16)	90.6% (1.38)	69.4% (1.36)	82.9% (1.73)	85.4% (1.83)
ABMRS ($\gamma = 1.2$)		95.9% (1.16)	92.1% (1.38)	66.3% (1.31)	83.0% (1.73)	83.9% (1.82)
TPRS	1000	97.8% (2.05)	96.7% (2.09)	94.3% (3.17)	95.7% (3.06)	96.2% (3.10)
TEPS		93.8% (0.69)	96.3% (1.97)	96.7% (2.78)	96.8% (2.93)	96.5% (3.05)
BMARS		87.2% (0.68)	84.7% (1.09)	65.2% (1.12)	79.7% (1.10)	56.3% (1.66)
GPR		97.3% (1.15)	96.5% (1.73)	97.0% (2.58)	96.6% (2.49)	97.3% (2.93)

* Values are reported as $a(b)$, where a denotes coverage and b denotes width ($\times 10^{-1}$).

Web Table 3: Additional results in Section 4.2.1: average coverage and width of 95% credible or confidence intervals for the mean function

	m	Surface 1	Surface 2	Surface 3
ABMRS ($\gamma = 0.8$)		83.8% (0.913)	87.9% (2.715)	89.2% (3.542)
ABMRS ($\gamma = 1.0$)		82.7% (0.901)	87.3% (2.688)	88.8% (3.502)
ABMRS ($\gamma = 1.2$)		82.5% (0.896)	85.9% (2.669)	88.7% (3.539)
TPRS	500	95.8% (1.195)	92.7% (5.164)	89.2% (5.086)
TEPS		96.8% (1.304)	92.0% (5.100)	90.6% (6.001)
BMARS		62.6% (0.703)	58.6% (3.193)	48.5% (3.134)
GPR		97.4% (1.185)	95.0% (5.587)	92.5% (6.209)
ABMRS ($\gamma = 0.8$)		82.5% (0.691)	92.4% (1.832)	93.0% (2.228)
ABMRS ($\gamma = 1.0$)		80.8% (0.677)	91.8% (1.840)	92.5% (2.215)
ABMRS ($\gamma = 1.2$)		80.2% (0.674)	92.1% (1.845)	92.4% (2.220)
TPRS	1000	96.7% (0.904)	92.1% (4.081)	88.1% (3.930)
TEPS		96.9% (0.949)	90.7% (6.001)	88.4% (4.319)
BMARS		61.4% (0.509)	60.3% (2.104)	51.7% (2.528)
GPR		95.0% (0.881)	95.4% (6.209)	92.7% (4.865)

* Values are reported as $a(b)$, where a denotes coverage and b denotes width.

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